

Assignment 3

Task 1:

Calculate the average public expenditure for the year 2005 across all countries.

Sol:

Step 1: Import the Datasets

- Open Power BI Desktop.
- Go to Home → Click Get Data → Choose Excel.
- Browse and select both files:
- Data.xlsx
- Expenditure.xlsx
- Select the relevant sheets from both files and click Load.

Step 2: Explore the Tables

- After loading:
- Review which table contains public expenditure by year.
- Look for columns like:
- Country
- Year
- Public Expenditure or similar field
- The 2005 data should appear either as a column or a row entry.

Step 3: Transform the Data

- Go to Transform Data (Power Query Editor).
- Select columns with year headers (e.g., 2000, 2001, ..., 2020).
- Right-click → Choose Unpivot Columns.
- Rename:
- Attribute → Year
- Value → PublicExpenditure

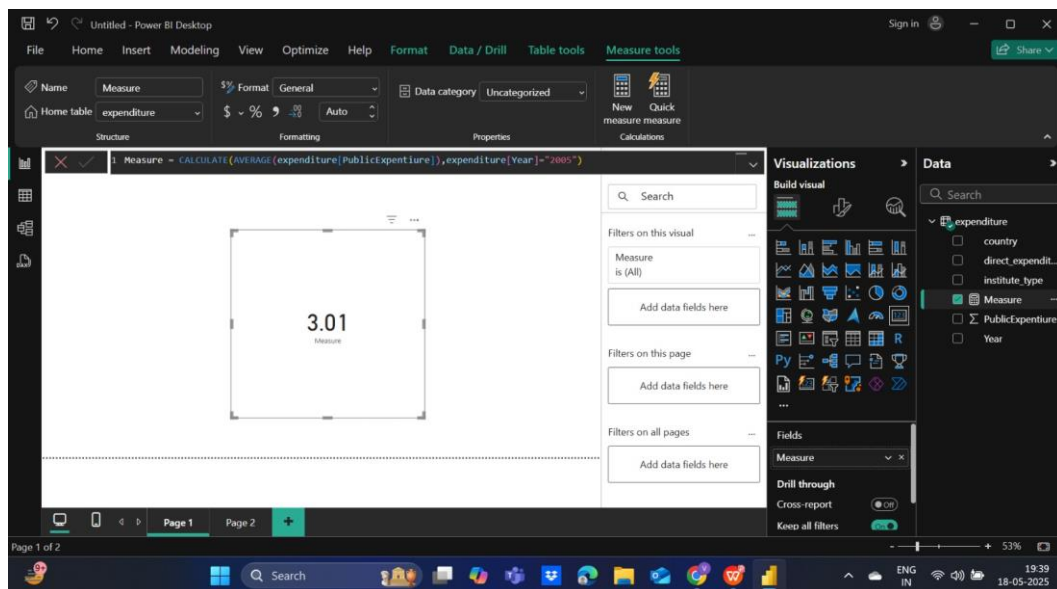
Step 4: Filter to Year 2005 and Calculate Average

- In the Fields pane, drag the Year field into the Filters pane (visual or page-level).
- Filter to 2005.

- Insert a Card visual:
- Drag the PublicExpenditure field into the card.
- From the dropdown, choose Average (not Sum or Count).

Final Output:

We should now have a single number displayed on a Card visual — the average public expenditure in 2005 across all countries.



Task 2:

Calculate the total publications for institutions in the UK.

Sol:

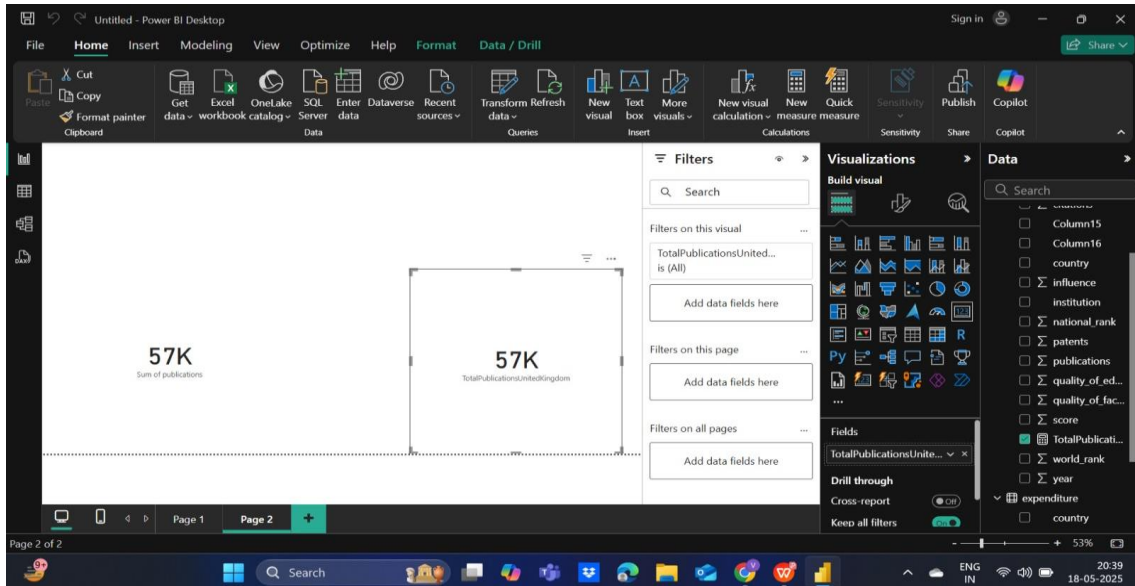
Step 1: Load the Dataset

- Open Power BI Desktop.
- Click on Home > Get Data > Excel.
- Select your Data.xlsx file.
- In the Navigator, check the sheet that contains publication data.
- Click Load.

Step 2: Create a Measure

- Go to the Modeling tab.
- Click New measure.

DAX Formula: TotalPublicationsUK = CALCULATE(SUM(Data[Publications]), Data[Country] = "United Kingdom")



Task 3:

Filter the dataset to show only institutions with a world rank below 100.

Sol:

Step 1: Load Your Data

- Make sure your dataset (e.g., Data.xlsx) is already loaded into Power BI.
- Ensure there is a column called something like World Rank, Rank, or similar.

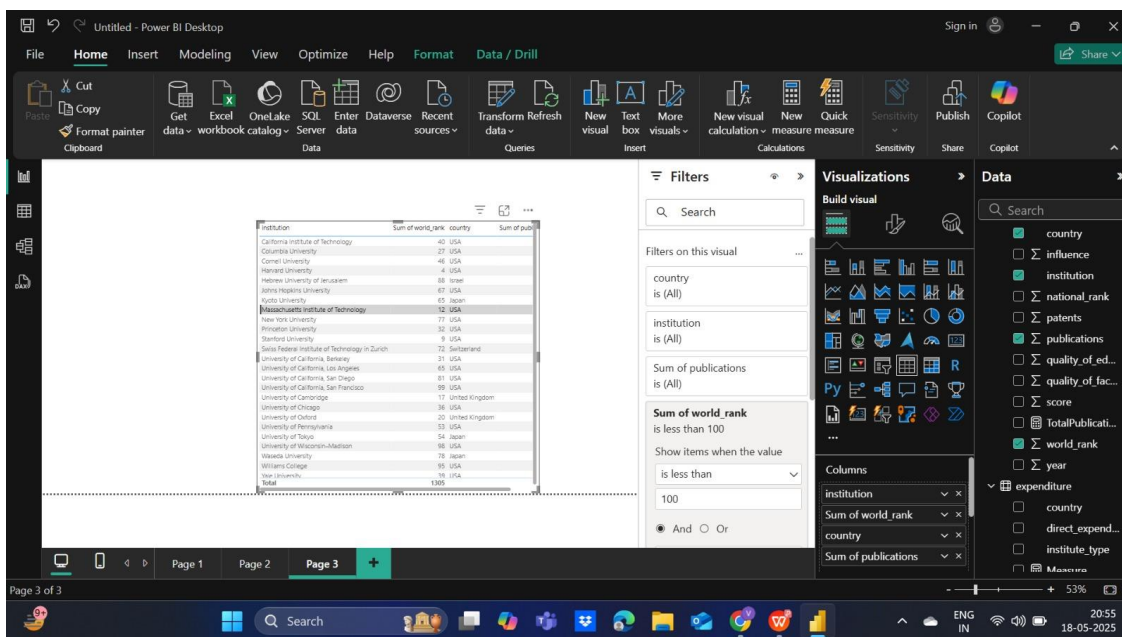
Step 2: Add a Table Visual (Optional but Helpful)

- Go to Report View (left pane with chart icon).
- Click on the Table visual in the Visualizations pane.
- Drag in fields like:
 - Institution Name
 - World Rank
 - Any other relevant info (like Country, Publications, etc.)

Step 3: Apply the Filter for World Rank < 100

- Select the visual (e.g., table or chart) that you want to filter.

- Go to the Filters pane (on the right side).
- Find the World Rank field in the Filters section.
- Drag the World Rank field to the Filters on this visual section.
- Change the filter type to "Advanced filtering".
- In the box, set:
- "is less than" → 100
- Click Apply filter.



Task 4:

Calculate the total expenditure for all years for each country.

Sol:

Step 1: Load the Data

- Open Power BI Desktop.
- Click Home > Get Data > Excel.
- Select Expenditure.xlsx and import the relevant sheet (e.g., Sheet1 or Expenditure).
- Click Load.

Step 2: Verify Data Columns

- Go to the Data View (left pane with table icon) and confirm:
- Country column is text
- Year column is whole number
- PublicExpenditure column is numeric (decimal or whole).

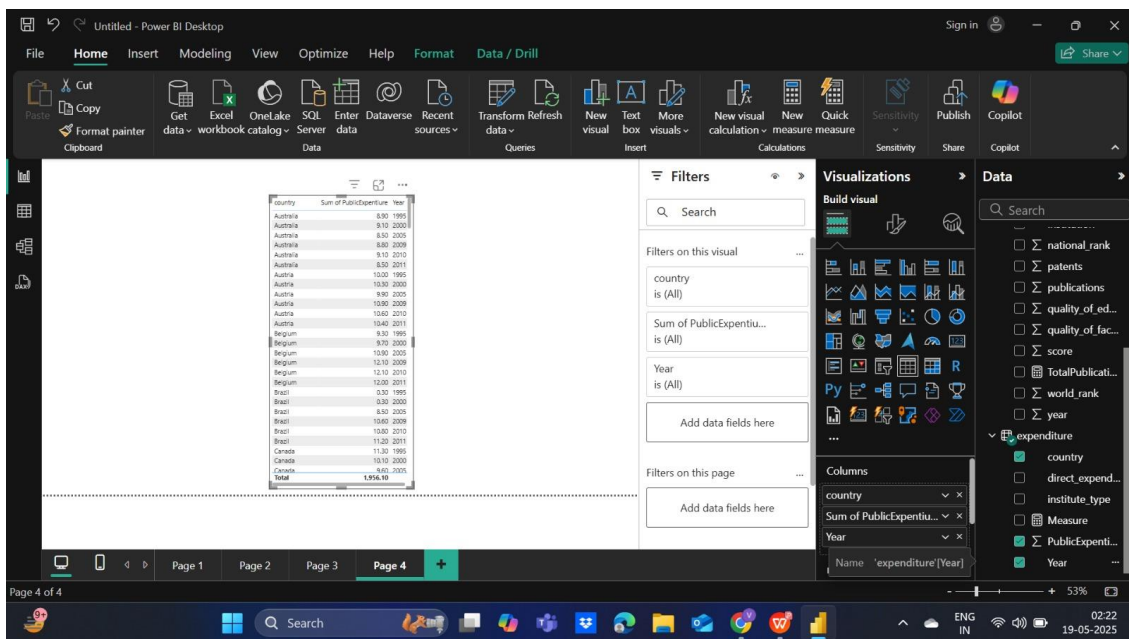
Step 3: Create a Visual to Show the Total

- Go to Report View (left pane with chart icon).
- Click on the Table visual from the Visualizations pane.
- Drag the following fields into the Values area:
- Country
- PublicExpenditure
- The visual will automatically group by country and sum expenditures across all years.

DAX Formula: TotalExpenditure = SUM(Expenditure[PublicExpenditure])

Step 4: Add a Filter or Slicer for Year

- Insert a Slicer visual.
- Drag the Year field into the slicer.
- Now users can choose specific years, but the default table still shows totals across all years unless filtered.



Task 5:

Write a DAX formula to ignore any filters on the year column and calculate the total score across all years.

Sol:

Step 1: Open Power BI and Load Your Data

- Ensure your dataset is already loaded and has:
- A column called Score
- A column called Year

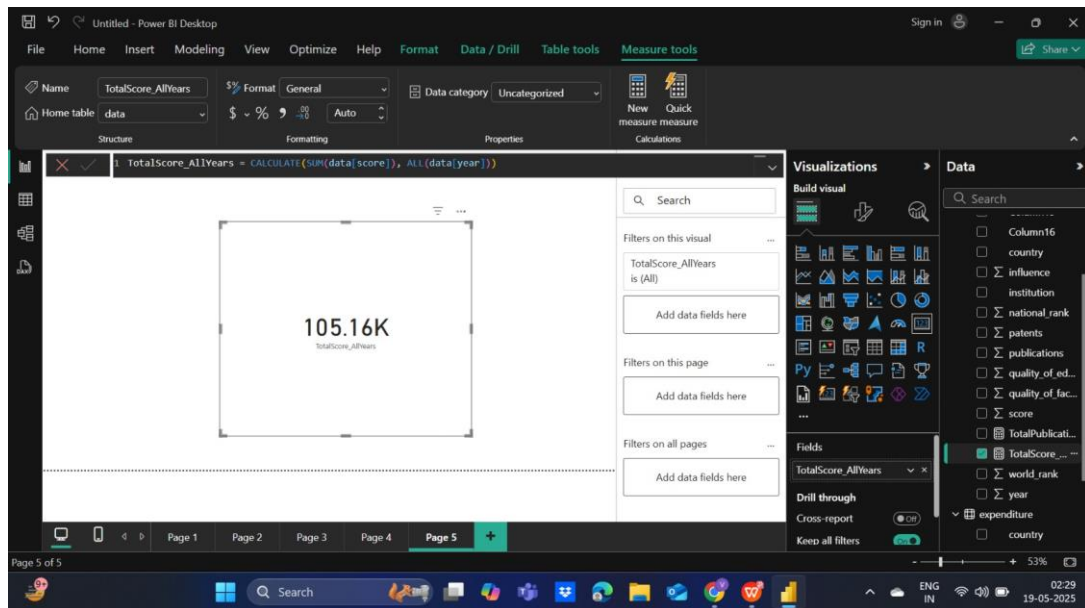
Step 2: Create a New Measure

- Go to the Modeling tab at the top.
- Click on New measure.

DAX FORMULA: TotalScore_AllYears =
CALCULATE(SUM(YourTableName[Score]), ALL(YourTableName[Year]))

Step 3: Use the Measure in a Visual

- Insert a Card or Table visual.
- Drag the measure TotalScore_AllYears into the visual.
- Even if you have a Year slicer or filter applied, this measure will still show the total score for all years, because the ALL() function removes the Year filter.



Task 6:

Calculate the growth in expenditure for Austria from 1995 to 2000.

Sol:

Step 1: Load the Data

- Open Power BI.
- Go to Home > Get Data > Excel.
- Select the Expenditure.xlsx file.
- Choose the correct sheet and click Load.
- Make sure the data contains:
- A column named Country
- A column named Year
- A column named PublicExpenditure (or similar)

Step 2: Filter for Austria (Optional, for testing)

- You can first filter the data to show just Austria's values in a Table visual to verify:
- Add a Table visual to the report.
- Drag Country, Year, and PublicExpenditure into the table.
- Add a slicer for Country, select Austria.
- This helps you confirm that 1995 and 2000 data for Austria exist.

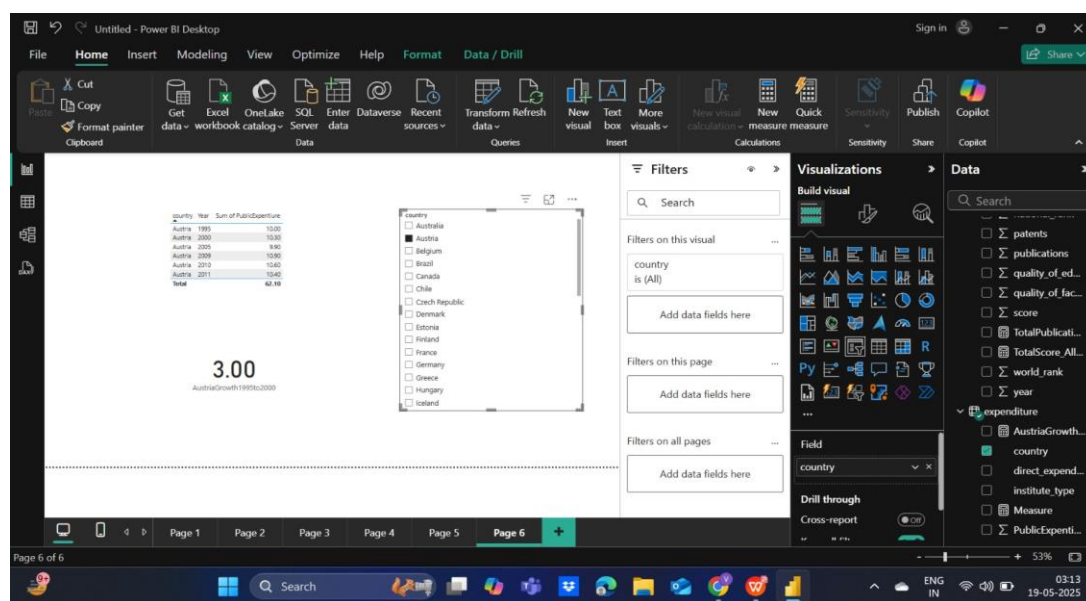
Step 3: Create a Measure to Calculate Growth

Go to the Modeling tab → click New Measure.

Dax Formula: AustriaGrowth1995to2000 = VAR Expenditure1995 = CALCULATE(SUM(Expenditure[PublicExpenditure]), Expenditure[Country] = "Austria", Expenditure[Year] = 1995) VAR Expenditure2000 = CALCULATE(SUM(Expenditure[PublicExpenditure]), Expenditure[Country] = "Austria", Expenditure[Year] = 2000) RETURN DIVIDE(Expenditure2000 - Expenditure1995, Expenditure1995) * 100.

Step 4: Display the Result

- Add a Card visual.
- Drag the AustriaGrowth1995to2000 measure into the card.
- The card will show the % growth in expenditure from 1995 to 2000 for Austria.



Task 7:

Format the expenditure values to include a currency symbol and zero decimal places.

Sol:

Step 1: Open Data View

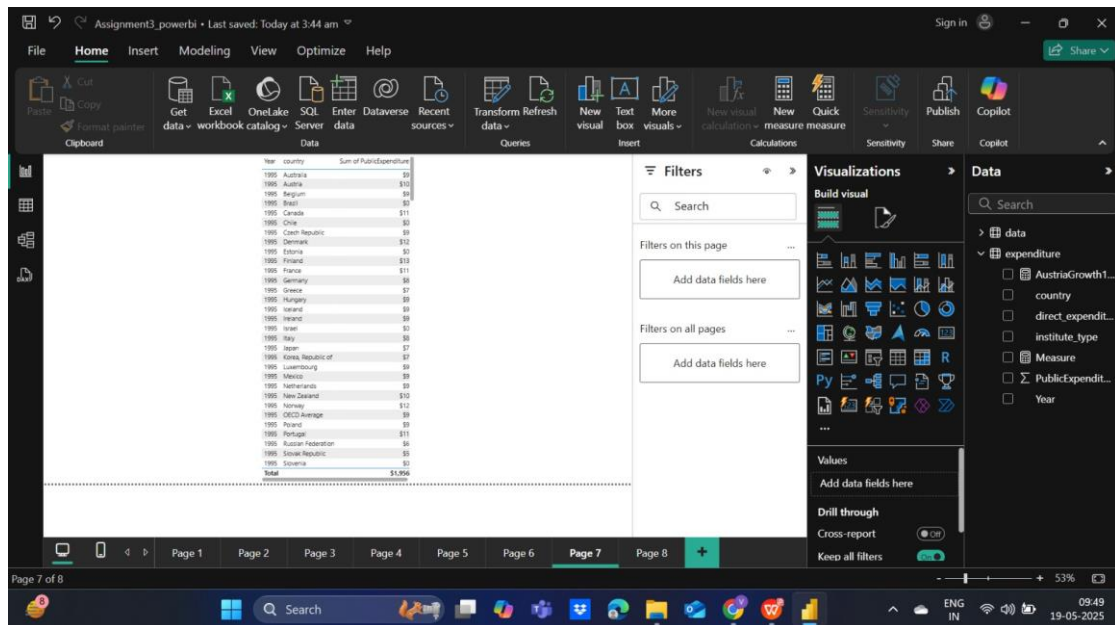
- In Power BI Desktop, click the Data view icon .
- Click the table (e.g., Expenditure) where your column is located.

Step 2: Select the Column

- Click on the column you want to format — for example, PublicExpenditure.

Step 3: Format the Column

- Go to the Modeling tab on the top ribbon.
- Look for the Formatting group.
- Under Data type, make sure it's set to Decimal number or Whole number.
- Under Format, do the following:
 - From the Format dropdown, choose Currency.
 - Set Decimal places to 0.



Task 8:

Calculate the total expenditure for each country.

Sol:

Step 1: Load the Data

- Open Power BI Desktop.
- Click Home > Get Data > Excel.
- Select the file: Expenditure.xlsx.
- Choose the sheet or table (usually named something like Expenditure) and click Load.

Step 2: Check Your Fields

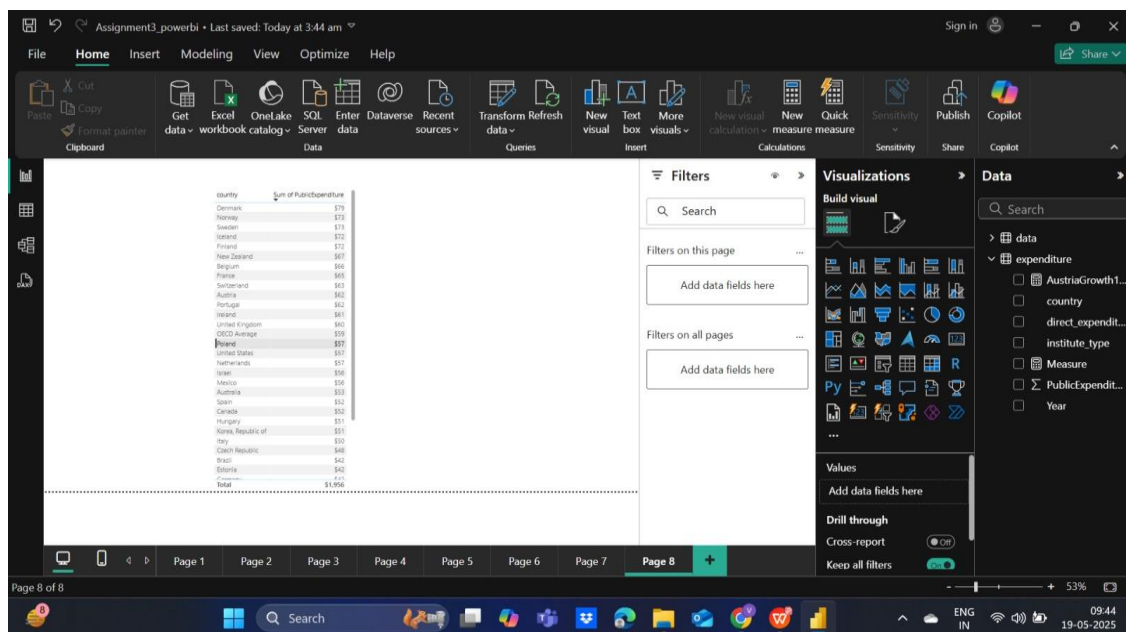
- Make sure these fields are available in the Fields pane:
- Country
- Year
- PublicExpenditure (or similar)

Step 3: Create a Visual to Display Total Expenditure by Country

- Option 1: Use a Table visual
- In the Visualizations pane, click the Table icon.
- Drag Country to the Rows/Values area.
- Drag PublicExpenditure to the Values area.
- Power BI will automatically sum the values.
- Result: A table showing each country with its total expenditure across all years.

Step 4: Format the Values (Optional)

- Click on the PublicExpenditure field in the Fields pane.
- Go to the Modeling tab (if visible) or use the Properties pane.
- Set Data Type to Decimal Number and Format to Currency with 0 decimal places.



Task 9:

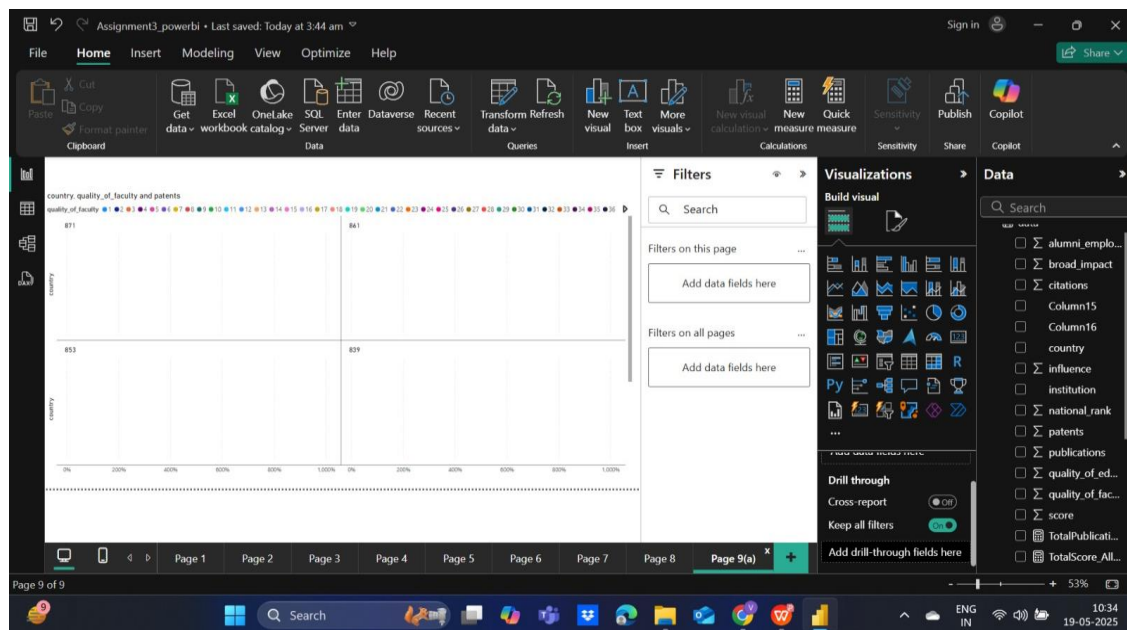
Create a report showing each visual in different sheets:

Part(a)

Break down the total patents of institutions by country and then by quality of faculty. Analyze which factors contribute most to the number of patents across different countries.

Sol:

- Go to New Page.
- Insert a clustered bar chart (or stacked column if you prefer).
- In the Fields pane, drag:
 - country to the Axis.
 - quality_of_faculty to Legend.
 - patents to Values (aggregation: Sum).
- Sort by patents descending.
- Add Data Labels from the Format pane for clarity.
- Analyze which faculty quality levels are associated with the highest patent outputs in which countries.

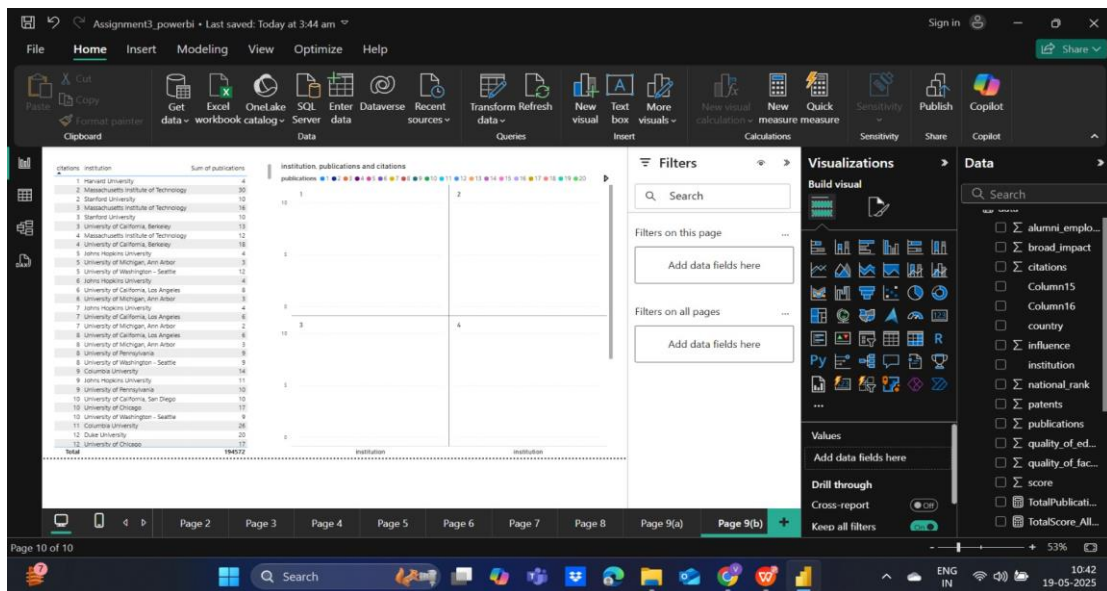


Part(b)

Use the Q&A feature in Power BI to answer the question: "What is the total publications and citations for institutions in the USA?" and display the results in a table and bar chart format.

Sol:

- Go to Page 2.
- Click on the Q&A visual from the Visualizations pane.
- Type:
- "Total publications and citations for institutions in the USA"
- Power BI should show a table. If not:
- Manually filter the country = USA in a Table visual.
- Drag institution, publications, and citations to the table.
- Create a bar chart:
- Axis: institution
- Values: publications or citations
- Use Legend if needed to separate them.



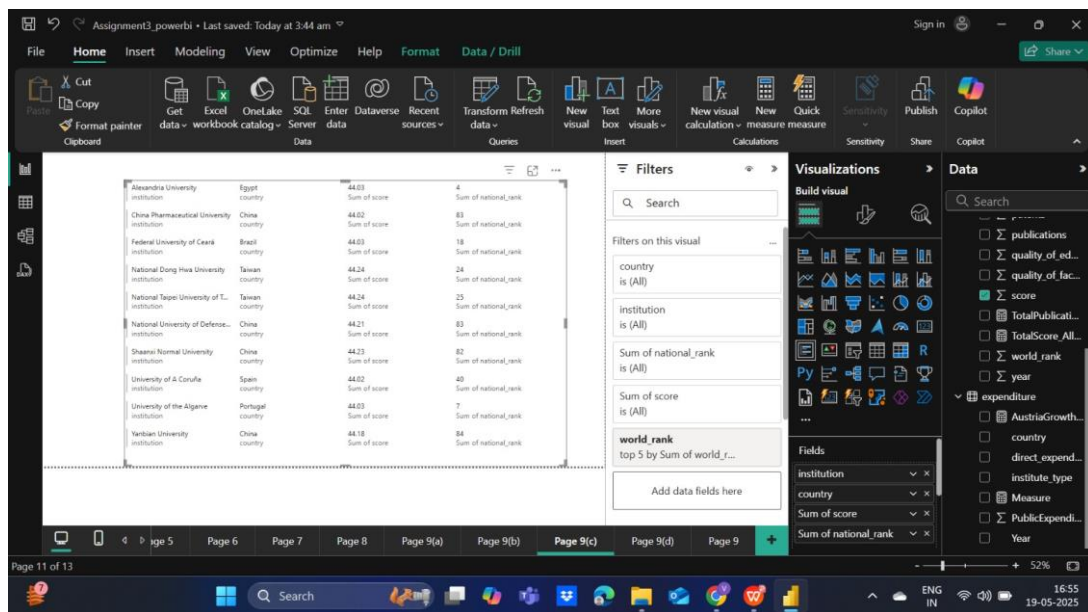
Part (c)

Display key metrics for the top 5 institutions by world rank, including fields such as institution, country, score, and national rank.

Sol:

- Go to Page 3.
- Create a Table or Multi-row Card visual.
- Drag:
- institution, country, score, and national_rank to the visual.
- In the Filters pane:
- Drag world_rank to Filters.
- Set to Top N → Enter "5" → Based on world_rank ascending.

- You can also sort the table by world rank directly.

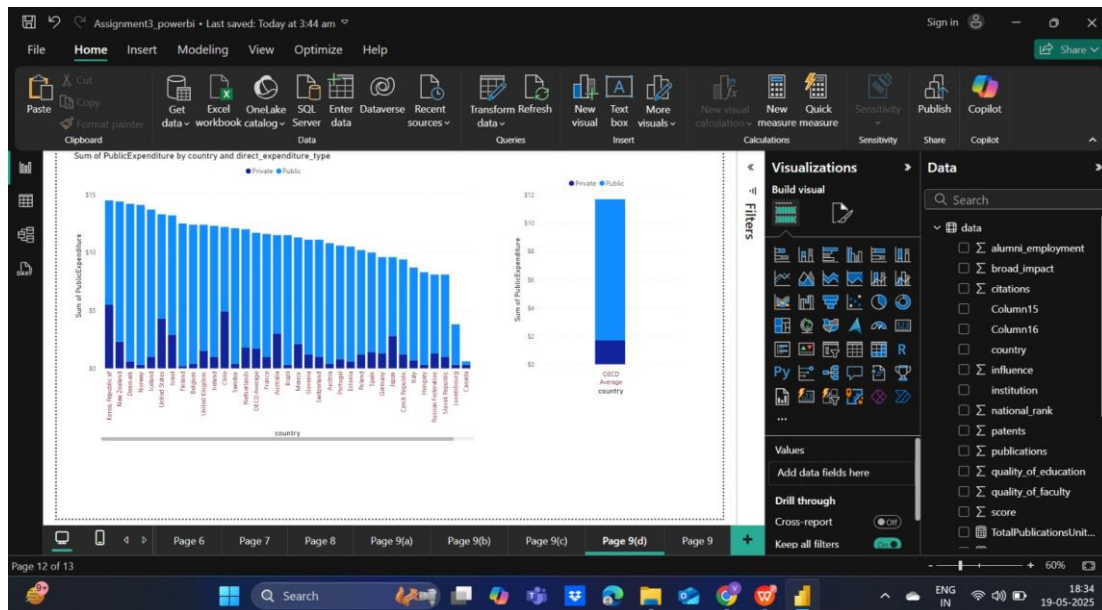


Part (d)

Represent the distribution of `direct_expenditure_type` (e.g., public vs. private) for the year 2011 across all countries. Highlight the OECD Average as a separate segment.

Sol:

- Go to Page 4.
- Use a Stacked column chart or 100% stacked bar chart.
- Drag:
 - country to Axis.
 - direct_expenditure_type to Legend.
 - PublicExpenditure to Values.
- Apply a filter:
 - In Filters pane, set Year = 2011.
- Highlight OECD Average:
 - Add Data label or use a separate card visual for OECD Average.
 - Optionally, format its color differently using DAX (like `IS_OECD = IF(country="OECD Average", "Yes", "No")`) and apply to conditional formatting.



Task 10:

Create a workspace "Institution Analysis" and set up a schedule to refresh the datasets every day at 6 AM.

Sol:

Step 1: Publish Your Report to Power BI Service

- Click File → Publish → To Power BI.
- Sign in with your Power BI account (if prompted).
- Click Create new workspace if needed, or publish to "My workspace" first.
- Choose the destination workspace or create a new one (next step).

Step 2: Create Workspace "Institution Analysis"

- Go to <https://app.powerbi.com>.
- In the left panel, click Workspaces.
- Click "+ Create a workspace".
- Enter:
- Name: Institution Analysis
- Optionally add a description or image.
- Click Save.

Step 3: Upload Your Report to the Workspace

- Inside the Institution Analysis workspace:

- Click “New” → Upload a file.
- Choose Local File and select your .pbix file.
- It will import the dataset and the report.

Step 4:

- In the Institution Analysis workspace,
- Click the “...” (More options) next to the dataset (not the report).
- Choose Settings.
- Expand the Scheduled refresh section.
- Set Keep data updated → On.
- Under Refresh frequency:
- Choose Daily.
- Under Time, set the time to 6:00 AM.
- You can add multiple times if needed.
- Under Data source credentials, click Edit credentials.
- Choose Authentication method (usually “OAuth2” or “Windows” for local files or gateways).
- Click Sign in to connect.
- Click **Apply** to save your schedule.

